

Systems Friction as a First-Class Design Requirement for Outsiders in Civic Digital Services

Independent research note (portfolio writeup · not peer-reviewed)

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Independent personal research ... not IU-endorsed or lab output

Abstract

Civic and public digital services often assume the user already has local context, language fluency, and unwritten process knowledge. For outsiders ... non-native speakers, new arrivals, and first-time users ... that assumption shows up as systems friction: stalled forms, PDF-only “language support,” blaming error messages, and channels that do not match outsider schedules. This independent research note locks a falsifiable claim that treating such friction as a first-class design requirement, rather than a post-launch accessibility fix, supports more accountable digital public services and public-interest AI adoption. Method: reflective pattern extraction from a Spring 2026 semester abroad in Madrid (Complutense / IU Overseas Study), documented as a public-safe pattern library, plus a reusable Outsider Impact Review one-pager for builders. Contribution at this stage is the claim, the patterns, the artifact template, and explicit limitations (single observer, one city, no controlled trial). No quantitative validation is claimed; a minimal kill test is specified for later work. This is a portfolio-style writeup, not a peer-reviewed publication, and it is not IU or lab research output.

1 Introduction

Public services increasingly run through forms, apps, and portals. When those systems are designed for people who already know the local script ... which office comes first, which document format is accepted, which language completes the task ... newcomers experience the same interface as a wall. That wall is easy to mis-read as personal failure. This note treats it as a design signal.

I spent Spring 2026 living and studying in Madrid through IU Overseas Study at Universidad Complutense. Coursework supplied vocabulary for institutional design; daily life supplied ground truth for how civic, university, commercial, and transit systems meet someone who does not yet belong. Parallel clinic work at ServeIT made the same failure modes visible in non-profit handoffs: tools built for the maintainer’s mental model, not the visitor’s first visit.

Existing accessibility practice (for example WCAG baselines used in clinic work) catches many markup and interaction failures, but a happy-path or checklist pass can still miss outsider-specific assumptions about local context, sequencing, and language-as-access. Closely related discussions of migrant administrative literacy, language barriers in e-government, and linguistic fairness in AI document related harms; the gap this note targets is narrower: a builder-facing requirement that outsider

friction gate automation and public digital changes before launch, not only as a late translation or accessibility patch.

This writeup’s contributions, stated as claims:

- Systems friction in civic digital services systematically disadvantages outsiders; the five patterns below are the working library from Madrid notes (suggestive, not causal proof).
- Treating that friction as a first-class design input ... via a reusable Outsider Impact Review ... is a practice bridge toward more accountable public digital services and public-interest AI adoption.
- The current method is reflective practice with an explicit kill test; no controlled experiment results are claimed here.

Section 2 situates prior themes; Section 3 describes method and design principles; Section 4 reports observed patterns; Section 5 discusses limits; Section 6 concludes.

2 Related Work

Three clusters frame the gap without claiming a complete literature survey (deep reading remains in progress in the source repo’s seed list).

Accessibility and comprehensibility. Clinic practice uses WCAG 2.1 as a baseline for keyboard paths, contrast, and structure [1]. Language access and plain-language defaults sit adjacent to comprehensibility even when checklists focus on markup. That overlap is necessary but not sufficient when the primary task is only completable after hidden local prerequisites.

Language, literacy, and civic access. Public notes in this project treat language choice as access control for task completion: Spanish-only defaults, PDF-only “English support,” and mixed-language households that need structural lang marking rather than a single page default. Adjacent prior work on migrant administrative literacy and e-government language barriers documents related burdens; this note does not claim to replace that literature, only to reframe friction as a pre-automation design gate for builders.

Public-interest AI and accountable automation. Serve-AI / ServeIT clinic instincts (human review, honest handoffs, accessibility before automation) rhyme with outsider friction as a ship/no-ship input. Policy and fairness discussions of public-sector automated decision-making supply accountability vocabulary; they rarely supply a portable one-pager students can run before shipping clinic or civic automation. This note’s artifact is aimed at that operational gap.

3 Approach / System

Method now: reflective practice and pattern extraction from lived systems use during one semester abroad, recorded as public-safe notes (no private administrative detail). Exploration versus confirmation are kept separate in the research log; confirmation experiments are planned, not run. The primary artifact is an Outsider Impact Review one-pager: before shipping a civic flow or adding automation, review assumed local context, unwritten sequencing, language-as-access, hours and channels, recoverable errors, and automation risk (whether AI would amplify those gaps).

Practice bridge only: ServeIT / Serve-AI clinic work is a rhyme, not a relicensing of those materials. Serve-AI project materials are CC BY-SA 4.0 with project credit; this madrid writeup and repo remain independent personal notes under MIT.

3.1 Design Principles

- Plain-language and completable language paths for the primary task (not PDF-only “support”).
- Sequencing stated in the interface; no hidden prerequisite steps.
- Recoverable errors that name the fix and preserve state.
- Human escalation with case identity when the digital path fails.

- No assumed prior belonging: design for the first-time outsider, then add automation only if the access baseline works.

4 Findings / Evaluation

Patterns below appeared repeatedly in lived use of civic, university, commercial, and transit/utility systems in Madrid. They suggest design failure modes; they do not prove causation beyond this observer’s semester. Recording method: public-safe reflective notes and essay outlines in the madrid-ai-ethics repository.

1. Assumed local context. Forms and apps expect address history, bank relationships, or national ID workflows a newcomer has never used.
2. Unwritten sequencing. The flow works only if step B happened before step A, but the interface does not say so.
3. Language as access control. Spanish-only defaults; “English support” that is a PDF rather than a completable interface; language choice functions like a login wall for task completion.
4. Hours and channels that do not match outsider life. In-person only, narrow windows, little or no status tracking after submit ... especially hard for students and new arrivals.
5. Error messages that blame the user. “Invalid document” instead of naming the fix (for example a different scan format); recoverable error UX is missing when stakes are highest.

Same failure modes show up in ServeIT clinic observation: tools built for the maintainer’s mental model, not the visitor’s first visit. Evaluation bar for later work (not yet run): apply Outsider Impact Review and a parallel happy-path / WCAG-style pass to two public civic flows; if the outsider pass finds nothing the happy-path pass misses, the “first-class requirement” half of the claim is in serious trouble for this artifact. No baselines, sample sizes, or percentage improvements are claimed here.

5 Discussion

Limitations are part of the claim package. Single observer: selection and confirmation bias are real. One city and one semester (Madrid, Spring 2026): other municipalities and seasons may differ. Reflective method: pattern extraction from lived use, not lab measurement. Public-safe scope omits private details that might strengthen or nuance a finding. Related-work deep reading is still owned by the author; the seed list is not a finished survey. This note does not claim legal or policy expertise, statistical generalizability, or that automation always harms outsiders ... poorly scoped automation

can amplify gaps; that is a design risk, not a theorem. It is not IRB, clinical, or health-system research. Affiliation lines name where the author studies; they do not mean Indiana University, ServeIT Clinic, or any lab endorses or owns this content. Readers should not treat this PDF as a published conference paper or as official IU research.

6 Conclusion

Outsiders hit civic digital systems where design assumes belonging. This independent research note locks that friction as a first-class design concern, documents five public-safe patterns from Madrid reflective practice, and offers an Outsider Impact Review template for builders before automation. The next empirical step is the dual-review kill test ... not invented metrics. Until that runs, the contribution is the claim, the patterns, the artifact, and the limits.

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References

- [1] W3C Web Accessibility Initiative, Web Content Accessibility Guidelines (WCAG) 2.1, W3C Recommendation, 5 June 2018. <https://www.w3.org/TR/WCAG21/>.
- [2] ServeIT Clinic, Luddy School of Informatics, Computing, and Engineering, Indiana University. <https://serveit.luddy.indiana.edu/>.
- [3] Serve-AI project page (ServeIT Clinic / PIT-UN track). <https://serveit.luddy.indiana.edu/serve-ai/index.html>.